

PREMIER UNIVERSITY CHITTAGONG

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING



Report On

Analyze the IMDB Movies dataset using Hadoop MapReduce and Apache Pig

Course Code: CSE 4346

Course Title: Big Data Analysis Laboratory

Submitted by

Mohammad Ashfak

ID: 0222220005101179

Section: C

Batch: 42nd

Session: Spring 2026

SUBMITTED TO:

MD Tamim Hossain

Lecturer,

Department of CSE,

Premier University Chittagong

September, 2026

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1. Problem Statement

Analyze the IMDB Movies dataset (imdbmovies.csv) using Hadoop MapReduce and Apache Pig to extract insights such as top-rated movies, genre popularity, prolific directors, and rating trends across decades.

1.1 Dataset Description

Column (in file)	Type	Description
Title_ID	string	Unique IMDb-style ID (e.g. tt0000001). 2,000,000 unique values, no nulls.
Title	string	Title of the movie. 1,596,761 unique values; 7 missing.
Title_Type	string	Category of title: short, movie, video, tvMovie, tvSpecial, videoGame.
Year	int	Release year. Range 0–2032; 0 appears to represent ‘unknown’. No nulls.
Genres	string	One or more genres, comma-separated, quoted when multiple. 2,111 unique combinations.
Runtime	int (not float)	Runtime in minutes. Stored as 0, not blank, when unknown (736,857 rows = 36.8%).
Rating	float	IMDb-style rating, 0–10. Stored as 0.0, not blank, when unrated. Median value is 0.
Votes	int	Number of votes backing the rating.
Director_ID	string	IMDb-style director ID (e.g. nm0005690).
Director_Name	string	Director name(s), comma-separated if more than one. 2 missing values.

1.2 Summary of Data Analysis

Following analysis can be performed on the data-set:

1. Total Movies by Genre
2. Movie Count per Director
3. Top 10 Highest Rated Movies
4. Decade-wise Average Rating
5. Year-wise Title Count

Following Pig analysis is performed on the data-set:

1. Movie Count Per Genre
2. Top Directors by Movie Count
3. Top 10 Highest-Rated Movies
4. Average Rating per Genre
5. Average Rating & Release Count per Year
6. Content Count by Title Type
7. Bottom 20 Lowest-Rated Titles

2. Map Reduce Analysis

2.1 Job 1: Total Movies by Genre

The total number of titles is calculated for each genre to determine the distribution of movies across different genres.

Driver Class: (GenreDriver.java)

```
package genrecount;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class GenreDriver {

    public static void main(String[] args) throws Exception {
        if (args.length != 2) {
            System.err.println("Usage: GenreDriver <input> <output>");
            System.exit(-1);
        }

        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Genre Wise Title Count");
        job.setJarByClass(GenreDriver.class);

        // Mapper
        job.setMapperClass(GenreMapper.class);

        // Reducer
        job.setReducerClass(GenreReducer.class);

        // Mapper output
        job.setMapOutputKeyClass(Text.class);
        job.setMapOutputValueClass(IntWritable.class);

        // Final output
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);

        // Input
        FileInputFormat.addInputPath(job, new Path(args[0]));

        // Output
        FileOutputFormat.setOutputPath(job, new Path(args[1]));
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Class: (GenreMapper.java)

```
package genrecount;

import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class GenreMapper
    extends Mapper<LongWritable, Text, Text, IntWritable> {
```

```

private final Text genreKey = new Text();
private final IntWritable one = new IntWritable(1);

@Override
public void map(LongWritable key, Text value, Context context)
throws IOException, InterruptedException {
    String line = value.toString();
    // Skip header
    if (line.startsWith("Title_ID")) {
        return;
    }
    String[] fields = line.split("\t", -1);
    // Need at least 5 columns
    if (fields.length < 5) {
        return;
    }
    // Column 4 = Genres
    String genres = fields[4];
    if (genres.equals("Unknown") || genres.equals("\\\\N")) {
        return;
    }
    // Example: Action,Adventure,Drama
    String[] genreList = genres.split(",");
    for (String genre : genreList) {
        genre = genre.trim();
        if (!genre.isEmpty()) {
            genreKey.set(genre);
            context.write(genreKey, one);
        }
    }
}

```

Reducer Class: (GenreReducer.java)

```

package genrecount;
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class GenreReducer
extends Reducer<Text, IntWritable, Text, IntWritable> {
    private final IntWritable result = new IntWritable();

    @Override
    public void reduce(Text key, Iterable<IntWritable> values,
    Context context)
    throws IOException, InterruptedException {
        int total = 0;
        for (IntWritable value : values) {
            total += value.get();
        }
        result.set(total);
        context.write(key, result);
    }
}

```

EXECUTION

```
C:\Ashfak\IMDBMoviesAnalysis\genrecount>jar -cvf genrecount.jar -C classes .
added manifest
adding: genrecount/(in = 0) (out= 0)(stored 0%)
adding: genrecount/GenreDriver.class(in = 1656) (out= 901)(deflated 45%)
adding: genrecount/GenreMapper.class(in = 2214) (out= 994)(deflated 55%)
adding: genrecount/GenreReducer.class(in = 1703) (out= 719)(deflated 57%)

C:\Ashfak\IMDBMoviesAnalysis\genrecount>dir genrecount.jar
Volume in drive C has no label.
Volume Serial Number is 66AE-5995

Directory of C:\Ashfak\IMDBMoviesAnalysis\genrecount

09/13/2026  05:11 PM                3,500 genrecount.jar
               1 File(s)                3,500 bytes
               0 Dir(s)          190,767,312,896 bytes free

C:\Ashfak\IMDBMoviesAnalysis\genrecount>hdfs dfs -rm -r -f /imdb_movies/output/genrecount

C:\Ashfak\IMDBMoviesAnalysis\genrecount>hadoop jar genrecount.jar genrecount.GenreDriver /imdb_movies/input/imdb_movies.txt /imdb_movies/output/genrecount
2026-09-13 17:13:12,487 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 17:13:14,243 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-13 17:13:14,302 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Sudipta/.staging/job_1789296062547_0001
2026-09-13 17:13:14,962 INFO input.FileInputFormat: Total input files to process : 1
2026-09-13 17:13:15,157 INFO mapreduce.JobSubmitter: number of splits:2
2026-09-13 17:13:15,519 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789296062547_0001
2026-09-13 17:13:15,525 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-13 17:13:15,970 INFO conf.Configuration: resource-types.xml not found
2026-09-13 17:13:15,970 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-13 17:13:16,002 INFO impl.YarnClientImpl: Submitted application application_1789296062547_0001
2026-09-13 17:13:16,895 INFO mapreduce.Job: The url to track the job: http://DESKTOP-AD7D20Q:8088/proxy/application_1789296062547_0001/
2026-09-13 17:13:16,895 INFO mapreduce.Job: Running job: job_1789296062547_0001
2026-09-13 17:13:37,410 INFO mapreduce.Job: Job job_1789296062547_0001 running in uber mode : false
2026-09-13 17:13:37,410 INFO mapreduce.Job:  map 0% reduce 0%
2026-09-13 17:13:55,140 INFO mapreduce.Job:  map 50% reduce 0%
2026-09-13 17:14:01,347 INFO mapreduce.Job:  map 100% reduce 0%
2026-09-13 17:14:13,556 INFO mapreduce.Job:  map 100% reduce 100%
2026-09-13 17:14:13,600 INFO mapreduce.Job: Job job_1789296062547_0001 completed successfully
2026-09-13 17:14:13,840 INFO mapreduce.Job: Counters: 55
```

Execution

```
Killed map tasks=1
Launched map tasks=3
Launched reduce tasks=1
Data-local map tasks=3
Total time spent by all maps in occupied slots (ms)=40356
Total time spent by all reduces in occupied slots (ms)=14483
Total time spent by all map tasks (ms)=40356
Total time spent by all reduce tasks (ms)=14483
Total vcore-milliseconds taken by all map tasks=40356
Total vcore-milliseconds taken by all reduce tasks=14483
Total megabyte-milliseconds taken by all map tasks=41324544
Total megabyte-milliseconds taken by all reduce tasks=14830592

Map-Reduce Framework
  Map input records=2000001
  Map output records=3317381
  Map output bytes=37363938
  Map output materialized bytes=43998712
  Input split bytes=240
  Combine input records=0
  Combine output records=0
  Reduce input groups=28
  Reduce shuffle bytes=43998712
  Reduce input records=3317381
  Reduce output records=28
  Spilled Records=6634762
  Shuffled Maps =2
  Failed Shuffles=0
  Merged Map outputs=2
  GC time elapsed (ms)=566
  CPU time spent (ms)=19000
  Physical memory (bytes) snapshot=1111920640
  Virtual memory (bytes) snapshot=1416998912
  Total committed heap usage (bytes)=909639680
  Peak Map Physical memory (bytes)=467787776
  Peak Map Virtual memory (bytes)=569204736
  Peak Reduce Physical memory (bytes)=220909568
  Peak Reduce Virtual memory (bytes)=289599488

Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0

File Input Format Counters
  Bytes Read=172940614

File Output Format Counters
  Bytes Written=389
```

Execution

OUTPUT

```
C:\Ashfak\IMDBMoviesAnalysis\genrecount>hdfs dfs -cat /imdb_movies/output/genrecount/part-r-00000
Action 104453
Adult 111093
Adventure 60896
Animation 79231
Biography 34998
Comedy 332549
Crime 65922
Documentary 341620
Drama 540219
Family 55728
Fantasy 53753
Film-Noir 877
Game-Show 1662
History 28841
Horror 105914
Music 76069
Musical 22786
Mystery 41586
News 9547
Reality-TV 8962
Romance 90446
Sci-Fi 46625
Short 951588
Sport 32764
Talk-Show 5642
Thriller 83960
War 14838
Western 14812

C:\Ashfak\IMDBMoviesAnalysis\genrecount>
```

Output

2.2 Job 2: Movie Count per Director

Driver Class: (DirectorDriver.java)

```
package directorcount;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class DirectorDriver {
    public static void main(String[] args) throws Exception {
        if (args.length != 2) {
            System.err.println("Usage: DirectorDriver <input> <output>");
            System.exit(-1);
        }
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Director Wise Title Count");
        job.setJarByClass(DirectorDriver.class);
        // Mapper
        job.setMapperClass(DirectorMapper.class);
        // Reducer
        job.setReducerClass(DirectorReducer.class);
        // Mapper output
        job.setMapOutputKeyClass(Text.class);
        job.setMapOutputValueClass(IntWritable.class);
        // Final output
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);
        // Input
        FileInputFormat.addInputPath(job, new Path(args[0]));
        // Output
        FileOutputFormat.setOutputPath(job, new Path(args[1]));
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Class: (DirectorMapper.java)

```
package directorcount;

import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class DirectorMapper
    extends Mapper<LongWritable, Text, Text, IntWritable> {
    private final Text directorKey = new Text();
    private final IntWritable one = new IntWritable(1);

    @Override
    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {
        String line = value.toString();
        // Skip header
    }
}
```

```

    if (line.startsWith("Title_ID")) {
        return;
    }
    String[] fields = line.split("\t", -1);
    // Need at least 10 columns
    if (fields.length < 10) {
        return;
    }
    // Column 9 = Director_Name
    String director = fields[9].trim();
    // Ignore unknown directors
    if (director.isEmpty()
        || director.equals("Unknown")
        || director.equals("\N")) {
        return;
    }
    directorKey.set(director);
    context.write(directorKey, one);
}
}

```

Reducer Class: (DirectorReducer.java)

```

package directorcount;
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class DirectorReducer
    extends Reducer<Text, IntWritable, Text, IntWritable> {
    private final IntWritable result = new IntWritable();
    @Override
    public void reduce(Text key, Iterable<IntWritable> values,
        Context context)
        throws IOException, InterruptedException {
        int total = 0;
        for (IntWritable value : values) {
            total += value.get();
        }
        result.set(total);
        context.write(key, result);
    }
}

```

EXECUTION

```
Select Administrator: Command Prompt
C:\Ashfaq\IMDBMoviesAnalysis\directorcount\hadoop jar directorcount.jar directorcount.DirectorDriver /imdb_movies/input/imdb_movies.txt /imdb_movies/output/directorcount
2026-09-13 17:26:27,467 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 17:26:29,099 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-13 17:26:29,142 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Sudipta/.staging/job_1789296062547_0002
2026-09-13 17:26:29,583 INFO input.FileInputFormat: Total input files to process : 1
2026-09-13 17:26:29,753 INFO mapreduce.JobSubmitter: number of splits:2
2026-09-13 17:26:30,044 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789296062547_0002
2026-09-13 17:26:30,050 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-13 17:26:30,427 INFO conf.Configuration: resource-types.xml not found
2026-09-13 17:26:30,427 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-13 17:26:30,569 INFO impl.YarnClientImpl: Submitted application application_1789296062547_0002
2026-09-13 17:26:30,680 INFO mapreduce.Job: The url to track the job: http://DESKTOP-AD7D20Q:8088/proxy/application_1789296062547_0002/
2026-09-13 17:26:30,694 INFO mapreduce.Job: Running job: job_1789296062547_0002
2026-09-13 17:26:47,634 INFO mapreduce.Job: Job job_1789296062547_0002 running in uber mode : false
2026-09-13 17:26:47,640 INFO mapreduce.Job: map 0% reduce 0%
2026-09-13 17:27:04,200 INFO mapreduce.Job: map 50% reduce 0%
2026-09-13 17:27:11,353 INFO mapreduce.Job: map 100% reduce 0%
2026-09-13 17:27:19,491 INFO mapreduce.Job: map 100% reduce 100%
2026-09-13 17:27:20,526 INFO mapreduce.Job: Job job_1789296062547_0002 completed successfully
2026-09-13 17:27:20,732 INFO mapreduce.Job: Counters: 55
File System Counters
  FILE: Number of bytes read=35639837
  FILE: Number of bytes written=71997021
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=172940854
  HDFS: Number of bytes written=9679491
  HDFS: Number of read operations=11
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Killed map tasks=1
  Launched map tasks=3
  Launched reduce tasks=1
  Data-local map tasks=3
  Total time spent by all maps in occupied slots (ms)=38515
  Total time spent by all reduces in occupied slots (ms)=12154
  Total time spent by all map tasks (ms)=38515
  Total time spent by all reduce tasks (ms)=12154
  Total vcore-milliseconds taken by all map tasks=38515
  Total vcore-milliseconds taken by all reduce tasks=12154
  Total megabyte-milliseconds taken by all map tasks=39439360
  Total megabyte-milliseconds taken by all reduce tasks=12445696
Map-Reduce Framework
  Map input records=2000001
  Map output records=1708822
  Map output bytes=32222187
  Map output materialized bytes=35639843
  Input split bytes=240
  Combine input records=0
  Combine output records=0
  Reduce input groups=569011
  Reduce shuffle bytes=35639843
  Reduce input records=1708822
  Reduce output records=569011
  Spilled Records=3417644
  Shuffled Maps =2
  Failed Shuffles=0
  Merged Map outputs=2
  GC time elapsed (ms)=552
  CPU time spent (ms)=21057
  Physical memory (bytes) snapshot=1168207872
  Virtual memory (bytes) snapshot=1414041600
  Total committed heap usage (bytes)=905969664
  Peak Map Physical memory (bytes)=501268480
  Peak Map Virtual memory (bytes)=568037376
  Peak Reduce Physical memory (bytes)=236494848
  Peak Reduce Virtual memory (bytes)=280936448
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=172940614
File Output Format Counters
  Bytes Written=9679491
```

Execution

```
Select Administrator: Command Prompt
Job Counters
  Killed map tasks=1
  Launched map tasks=3
  Launched reduce tasks=1
  Data-local map tasks=3
  Total time spent by all maps in occupied slots (ms)=38515
  Total time spent by all reduces in occupied slots (ms)=12154
  Total time spent by all map tasks (ms)=38515
  Total time spent by all reduce tasks (ms)=12154
  Total vcore-milliseconds taken by all map tasks=38515
  Total vcore-milliseconds taken by all reduce tasks=12154
  Total megabyte-milliseconds taken by all map tasks=39439360
  Total megabyte-milliseconds taken by all reduce tasks=12445696
Map-Reduce Framework
  Map input records=2000001
  Map output records=1708822
  Map output bytes=32222187
  Map output materialized bytes=35639843
  Input split bytes=240
  Combine input records=0
  Combine output records=0
  Reduce input groups=569011
  Reduce shuffle bytes=35639843
  Reduce input records=1708822
  Reduce output records=569011
  Spilled Records=3417644
  Shuffled Maps =2
  Failed Shuffles=0
  Merged Map outputs=2
  GC time elapsed (ms)=552
  CPU time spent (ms)=21057
  Physical memory (bytes) snapshot=1168207872
  Virtual memory (bytes) snapshot=1414041600
  Total committed heap usage (bytes)=905969664
  Peak Map Physical memory (bytes)=501268480
  Peak Map Virtual memory (bytes)=568037376
  Peak Reduce Physical memory (bytes)=236494848
  Peak Reduce Virtual memory (bytes)=280936448
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=172940614
File Output Format Counters
  Bytes Written=9679491
```

Execution

OUTPUT

C:\ Select Administrator: Command Prompt

Yoh Komaya	4	
Yohahn Ko	2	
Yohai Hakak	2	
Yohan Begyn	1	
Yohan Faure	3	
Yohan Guignard	3	
Yohan Kurian	4	
Yohan Laffort	1	
Yohan Leduc	3	
Yohan Mabit	1	
Yohan Manca	2	
Yohan Molles	1	
Yohan Riba Martinez	4	
Yohan Rozenblat	1	
Yohan Rozev	1	
Yohan Samel	2	
Yohan Shim	1	
Yohan Tison	1	
Yohan Ungar	6	
Yohana Desta	1	
Yohana Papa Onyango	1	
Yohanan Doron	1	
Yohanan Weller	7	
Yohance Brinson	2	
Yohane Dean Lengol	2	
Yohann Abdelnour	1	
Yohann Angelvy	1	
Yohann Auroux-Bernard	1	
Yohann Benedetti	2	
Yohann Boennec	2	
Yohann Charrin	1	
Yohann Compin	1	
Yohann Costedoat-Descouzeres	1	
Yohann Gloaguen	7	
Yohann Grignou	1	
Yohann Guglielmetti	1	
Yohann Iwandza	3	
Yohann Kouam	4	
Yohann Sfez	3	
Yohann Thiou	2	
Yohann Thiriet	1	
Yohanna Idha	1	
Yohanne Lamoulere	1	
Yohannes Feleke	1	
Yohannes Yamassee	1	
Yohay Hayat	1	
Yohei Ishibashi	2	
Yohei Ishihama	1	
Yohei Kameyama	1	

Output

Zlatko Geleski	3	
Zlatko Gotovac	1	
Zlatko Grgic	15	
Zlatko Kopljari	1	
Zlatko Lavanic	4	
Zlatko Milojevic	1	
Zlatko Pavlinic	7	
Zlatko Pavlinovic	1	
Zlatko Sacer	1	
Zlatko Seselj	1	
Zlatko Stojilovic	3	
Zlatko Sudovic	13	
Zlatko Sviben	3	
Zlatko Tomic	9	
Zlatko Topcic	1	
Zlatolin Donchev	1	
Zlatoslava Kryshatafovych	1	
Zmacforthewin	1	
Zmago Lenardic	1	
Zman	1	
Zmarai Kasi	1	
Zmaster	1	
Zmjkk	1	
Znere Grace	1	
Zo Bell	1	
Zo Fan	2	
Zo Tahiana Hariminoson	1	
Zo Zosak	15	
Zoa Zalazar	1	
Zoana Price	2	
Zobayer Raihan	2	
Zobi	1	
Zobunus	1	
ZodWriter	1	
Zodiac Fishgrease	1	
Zodwa Nyoni	1	
Zoe Aiano	2	
Zoe Alker	2	
Zoe Allen	1	
Zoe Andrikidis	3	
Zoe Banks	1	
Zoe Barcza	1	
Zoe Barnes	1	
Zoe Barthelemy	1	
Zoe Beck	2	
Zoe Bell	1	
Zoe Belle Gronner	1	
Zoe Beloff	7	
Zoe Berriat	7	

Output

2.3 Job 3: Top 10 Highest Rated Movies

Driver Class: (TopMoviesDriver.java)

```
package topdirectors;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.FloatWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class TopDirectorsDriver {

    public static void main(String[] args) throws Exception {

        if (args.length < 3) {
            System.err.println("Usage: TopDirectorsDriver <input> <tempDir> <output>");
            System.exit(-1);
        }

        // Job 1: Aggregate Average Rating per Director
        Configuration conf1 = new Configuration();
        Job job1 = Job.getInstance(conf1, "DirectorAverageRating");
        job1.setJarByClass(TopDirectorsDriver.class);
        job1.setMapperClass(DirectorAvgMapper.class);
        job1.setReducerClass(DirectorAvgReducer.class);
        job1.setOutputKeyClass(Text.class);
        job1.setOutputValueClass(FloatWritable.class);
        FileInputFormat.addInputPath(job1, new Path(args[0]));
        FileOutputFormat.setOutputPath(job1, new Path(args[1]));
        boolean job1Done = job1.waitForCompletion(true);
        if (job1Done) {

            // Job 2: Sort Descending & Limit to Top 10
            Configuration conf2 = new Configuration();
            Job job2 = Job.getInstance(conf2, "TopDirectorsSort");
            job2.setJarByClass(TopDirectorsDriver.class);
            job2.setMapperClass(DirectorSwapMapper.class);
            job2.setMapOutputKeyClass(FloatWritable.class);
            job2.setMapOutputValueClass(Text.class);
            job2.setSortComparatorClass(DescendingRatingComparator.class);
            job2.setReducerClass(TopNDirectorsReducer.class);
            job2.setOutputKeyClass(Text.class);
            job2.setOutputValueClass(FloatWritable.class);
            job2.setNumReduceTasks(1);
            FileInputFormat.addInputPath(job2, new Path(args[1]));
            FileOutputFormat.setOutputPath(job2, new Path(args[2]));
            System.exit(job2.waitForCompletion(true) ? 0 : 1);
        }
    }
}
```

Mapper Class: (TopMoviesMapper.java)

```
package topmovies;

import java.io.IOException;
import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;
```

```

public class TopMoviesMapper
    extends Mapper<Object, Text, DoubleWritable, Text> {
    private final DoubleWritable ratingKey = new DoubleWritable();
    private final Text titleValue = new Text();

    @Override
    public void map(Object key, Text value, Context context)
        throws IOException, InterruptedException {
        String line = value.toString();
        // Skip header
        if (line.startsWith("Title_ID")) {
            return;
        }
        String[] fields = line.split("\t", -1);
        // Need at least 8 columns
        if (fields.length < 8) {
            return;
        }
        String titleId = fields[0].trim();
        String title = fields[1].trim();
        String rating = fields[6].trim();
        String votes = fields[7].trim();
        if (rating.equals("\N") || rating.equals("Unknown")
            || title.equals("\N") || title.isEmpty()) {
            return;
        }
        try {
            double ratingValue = Double.parseDouble(rating);
            String output =
                titleId + "\t" + title + "\tRating=" + ratingValue
                + "\tVotes=" + votes;
            ratingKey.set(ratingValue);
            titleValue.set(output);
            context.write(ratingKey, titleValue);
        } catch (NumberFormatException e) {
            // Ignore invalid records
        }
    }
}

```

Reducer Class: (TopMoviesReducer.java)

```

package topmovies;

import java.io.IOException;
import java.util.PriorityQueue;
import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class TopMoviesReducer
    extends Reducer<DoubleWritable, Text, DoubleWritable, Text> {
    private PriorityQueue<MovieRecord> topMovies;
    private static final int TOP_N = 10;

    @Override
    protected void setup(Context context) {
        topMovies = new PriorityQueue<MovieRecord>(
            TOP_N,
            (a, b) -> Double.compare(a.rating, b.rating)
        );
    }
}

```

```

@Override
public void reduce(DoubleWritable key,
Iterable<Text> values,
Context context)
throws IOException, InterruptedException {
    for (Text value : values) {
        MovieRecord movie =
            new MovieRecord(key.get(), value.toString());
        topMovies.offer(movie);
        if (topMovies.size() > TOP_N) {
            topMovies.poll();
        }
    }
}

@Override
protected void cleanup(Context context)
throws IOException, InterruptedException {
    MovieRecord[] results =
        topMovies.toArray(new MovieRecord[0]);
    java.util.Arrays.sort(
        results,
        (a, b) -> Double.compare(b.rating, a.rating)
    );
    for (MovieRecord movie : results) {
        context.write(
            new DoubleWritable(movie.rating),
            new Text(movie.info)
        );
    }
}

private static class MovieRecord {
    double rating;
    String info;
    MovieRecord(double rating, String info) {
        this.rating = rating;
        this.info = info;
    }
}
}

```

EXECUTION

```
C:\Ashfak\IMDBMoviesAnalysis\topmovies>hadoop jar topmovies.jar topmovies.TopMoviesDriver /imdb_movies/input/imdb_movies.txt /imdb_movies/output/topmovies
2026-09-13 17:43:03,939 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 17:43:05,299 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-13 17:43:05,339 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Sudipta/.staging/job_1789296062547_0004
2026-09-13 17:43:05,778 INFO input.FileInputFormat: Total input files to process : 1
2026-09-13 17:43:05,964 INFO mapreduce.JobSubmitter: number of splits:2
2026-09-13 17:43:06,296 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789296062547_0004
2026-09-13 17:43:06,300 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-13 17:43:06,702 INFO conf.Configuration: resource-types.xml not found
2026-09-13 17:43:06,702 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-13 17:43:06,820 INFO impl.YarnClientImpl: Submitted application application_1789296062547_0004
2026-09-13 17:43:06,895 INFO mapreduce.Job: The url to track the job: http://DESKTOP-AD7D20Q:8088/proxy/application_1789296062547_0004/
2026-09-13 17:43:06,903 INFO mapreduce.Job: Running job: job_1789296062547_0004
2026-09-13 17:43:22,487 INFO mapreduce.Job: Job job_1789296062547_0004 running in uber mode : false
2026-09-13 17:43:22,492 INFO mapreduce.Job: map 0% reduce 0%
2026-09-13 17:43:41,285 INFO mapreduce.Job: map 50% reduce 0%
2026-09-13 17:43:45,435 INFO mapreduce.Job: map 83% reduce 0%
2026-09-13 17:43:51,673 INFO mapreduce.Job: map 100% reduce 0%
2026-09-13 17:43:59,810 INFO mapreduce.Job: map 100% reduce 100%
2026-09-13 17:44:00,837 INFO mapreduce.Job: Job job_1789296062547_0004 completed successfully
2026-09-13 17:44:01,073 INFO mapreduce.Job: Counters: 55
File System Counters
  FILE: Number of bytes read=212929979
  FILE: Number of bytes written=333658343
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=172940854
  HDFS: Number of bytes written=542
  HDFS: Number of read operations=11
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0
Job Counters
  Killed map tasks=1
  Launched map tasks=3
  Launched reduce tasks=1
  Data-local map tasks=3
  Total time spent by all maps in occupied slots (ms)=49478
  Total time spent by all reduces in occupied slots (ms)=16267
  Total time spent by all map tasks (ms)=49478
  Total time spent by all reduce tasks (ms)=16267
  Total vcore-millisecods taken by all map tasks=49478
  Total vcore-millisecods taken by all reduce tasks=16267
  Total megabyte-millisecods taken by all map tasks=50665472
  Total megabyte-millisecods taken by all reduce tasks=16657408
```

Execution

```
Map-Reduce Framework
  Map input records=2000001
  Map output records=2000000
  Map output bytes=116000485
  Map output materialized bytes=120011029
  Input split bytes=240
  Combine input records=0
  Combine output records=0
  Reduce input groups=92
  Reduce shuffle bytes=120011029
  Reduce input records=2000000
  Reduce output records=10
  Spilled Records=5551277
  Shuffled Maps =2
  Failed Shuffles=0
  Merged Map outputs=2
  GC time elapsed (ms)=616
  CPU time spent (ms)=26852
  Physical memory (bytes) snapshot=1325338624
  Virtual memory (bytes) snapshot=1577426944
  Total committed heap usage (bytes)=1088946176
  Peak Map Physical memory (bytes)=504123392
  Peak Map Virtual memory (bytes)=572047360
  Peak Reduce Physical memory (bytes)=355848192
  Peak Reduce Virtual memory (bytes)=433084032
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=172940814
File Output Format Counters
  Bytes Written=542
```

Execution

OUTPUT

```
C:\Ashfak\IMDBMoviesAnalysis\topmovies>hdfs dfs -cat /imdb_movies/output/topmovies/part-r-00000
10.0      tt34727714      The Clay Horse      Rating=10.0      Votes=26
10.0      tt21065190      Native Renewables: Powering a Nation with the Sun      Rating=10.0      Votes=9
10.0      tt29807669      English Rose      Rating=10.0      Votes=8
10.0      tt28510081      Eutanazija      Rating=10.0      Votes=8
10.0      tt2426280      Catador de sonhos      Rating=10.0      Votes=8
10.0      tt28669927      |Ebermensch      Rating=10.0      Votes=21
10.0      tt23962784      Internal      Rating=10.0      Votes=7
10.0      tt26941897      Bihari lives Matter      Rating=10.0      Votes=10
10.0      tt2955724      Losing Senses      Rating=10.0      Votes=19
10.0      tt3101358      MILFs Like It Dark      Rating=10.0      Votes=6

C:\Ashfak\IMDBMoviesAnalysis\topmovies>
```

Output

2.4 Job 4:Decade-Wise Average Rating

Driver Class: (decadeavg.java)

```
package decadeavg;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class DecadeDriver {

    public static void main(String[] args) throws Exception {

        if (args.length != 2) {
            System.err.println("Usage: DecadeDriver <input> <output>");
            System.exit(-1);
        }

        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Decade Wise Average Rating");
        job.setJarByClass(DecadeDriver.class);
        // Mapper
        job.setMapperClass(DecadeMapper.class);
        // Reducer
        job.setReducerClass(DecadeReducer.class);
        // Mapper output
        job.setMapOutputKeyClass(Text.class);
        job.setMapOutputValueClass(DoubleWritable.class);
        // Final output
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(DoubleWritable.class);
        // Input
        FileInputFormat.addInputPath(job, new Path(args[0]));
        // Output
        FileOutputFormat.setOutputPath(job, new Path(args[1]));
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Class: (DecadeMapper.java)

```
package decadeavg;

import java.io.IOException;
import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class DecadeMapper
    extends Mapper<LongWritable, Text, Text, DoubleWritable> {

    private final Text decadeKey = new Text();
    private final DoubleWritable ratingValue = new DoubleWritable();

    @Override
    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        // Skip header
```

```

    if (line.startsWith("Title_ID")) {
        return;
    }
    String[] fields = line.split("\t", -1);
    // Need at least 7 columns
    if (fields.length < 7) {
        return;
    }
    String year = fields[3].trim();
    String rating = fields[6].trim();
    if (year.equals("\\N") || year.equals("Unknown")
        || rating.equals("\\N") || rating.equals("Unknown")) {
        return;
    }
    try {
        int yearValue = Integer.parseInt(year);
        double ratingValueDouble = Double.parseDouble(rating);
        int decade = (yearValue / 10) * 10;
        decadeKey.set(decade + "s");
        ratingValue.set(ratingValueDouble);
        context.write(decadeKey, ratingValue);
    } catch (NumberFormatException e) {
        // Ignore invalid records
    }
}
}
}

```

Reducer Class: (DecadeReducer.java)

```

package decadeavg;
import java.io.IOException;
import org.apache.hadoop.io.DoubleWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class DecadeReducer
    extends Reducer<Text, DoubleWritable, Text, DoubleWritable> {
    private final DoubleWritable result = new DoubleWritable();
    @Override
    public void reduce(Text key, Iterable<DoubleWritable> values,
        Context context)
        throws IOException, InterruptedException {
        double sum = 0.0;
        int count = 0;
        for (DoubleWritable value : values) {
            sum += value.get();
            count++;
        }
        if (count > 0) {
            double average = sum / count;
            // Round to 2 decimal places
            average = Math.round(average * 100.0) / 100.0;
            result.set(average);
            context.write(key, result);
        }
    }
}

```

EXECUTION

Select Administrator: Command Prompt

```

C:\AshFak\IDDB\MoviesAnalysis\decadeavg>hadoop jar decadeavg.jar decadeavg.DecadeDriver /imdb_movies/input/imdb_movies.txt /imdb_movies/output/decadeavg
2026-09-13 17:36:49,128 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 17:36:50,631 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-13 17:36:50,675 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Sudipta/.staging/job_1789296062547_0003
2026-09-13 17:36:51,168 INFO input.FileInputFormat: Total input files to process : 1
2026-09-13 17:36:51,341 INFO mapreduce.JobSubmitter: number of splits:2
2026-09-13 17:36:51,700 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789296062547_0003
2026-09-13 17:36:51,702 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-13 17:36:52,132 INFO conf.Configuration: resource-types.xml not found
2026-09-13 17:36:52,148 INFO resource.ResourceUtilits: Unable to find 'resource-types.xml'.
2026-09-13 17:36:52,274 INFO impl.YarnClientImpl: Submitted application application_1789296062547_0003
2026-09-13 17:36:52,370 INFO mapreduce.Job: The url to track the job: http://DESKTOP-A07020Q:8888/proxy/application_1789296062547_0003/
2026-09-13 17:36:52,374 INFO mapreduce.Job: Running job: job_1789296062547_0003
2026-09-13 17:37:07,819 INFO mapreduce.Job: Job job_1789296062547_0003 running in uber mode : false
2026-09-13 17:37:07,819 INFO mapreduce.Job: map 0% reduce 0%
2026-09-13 17:37:22,335 INFO mapreduce.Job: map 50% reduce 0%
2026-09-13 17:37:25,388 INFO mapreduce.Job: map 100% reduce 0%
2026-09-13 17:37:35,561 INFO mapreduce.Job: map 100% reduce 100%
2026-09-13 17:37:35,577 INFO mapreduce.Job: Job job_1789296062547_0003 completed successfully
2026-09-13 17:37:35,779 INFO mapreduce.Job: Counters: 55
  File System Counters
    FILE: Number of bytes read=31628438
    FILE: Number of bytes written=63974196
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=172940854
    HDFS: Number of bytes written=191
    HDFS: Number of read operations=11
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=2
    HDFS: Number of bytes read erasure-coded=0
  Job Counters
    Killed map tasks=1
    Launched map tasks=2
    Launched reduce tasks=1
    Data-local map tasks=2
    Total time spent by all maps in occupied slots (ms)=27527
    Total time spent by all reduces in occupied slots (ms)=9924
    Total time spent by all map tasks (ms)=27527
    Total time spent by all reduce tasks (ms)=9924
    Total vcore-milliseconds taken by all map tasks=27527
    Total vcore-milliseconds taken by all reduce tasks=9924
    Total megabyte-milliseconds taken by all map tasks=28187648
    Total megabyte-milliseconds taken by all reduce tasks=10162176

```

Execution

Map-Reduce Framework

```

Map input records=2000001
Map output records=2000000
Map output bytes=27628432
Map output materialized bytes=31628444
Input split bytes=240
Combine input records=0
Combine output records=0
Reduce input groups=18
Reduce shuffle bytes=31628444
Reduce input records=2000000
Reduce output records=18
Spilled Records=4000000
Shuffled Maps =2
Failed Shuffles=0
Merged Map outputs=2
GC time elapsed (ms)=441
CPU time spent (ms)=15902
Physical memory (bytes) snapshot=1111613440
Virtual memory (bytes) snapshot=1368780800
Total committed heap usage (bytes)=868220928
Peak Map Physical memory (bytes)=500318208
Peak Map Virtual memory (bytes)=509430208
Peak Reduce Physical memory (bytes)=212361216
Peak Reduce Virtual memory (bytes)=270987264
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=172940814
File Output Format Counters
  Bytes Written=191

```

C:\AshFak\IDDB\MoviesAnalysis\decadeavg>

Execution

OUTPUT

```
C:\Ashfak\IMDBMoviesAnalysis\decadeavg>hdfs dfs -cat /imdb_movies/output/decadeavg/part-r-00000
3s      0.0
1870s    5.35
1880s    5.08
1890s    3.08
1900s    0.9
1910s    0.48
1920s    1.2
1930s    2.72
1940s    3.34
1950s    2.87
1960s    2.75
1970s    2.81
1980s    2.91
1990s    2.75
2000s    2.55
2010s    1.66
2020s    1.45
2030s    0.0
```

Output

2.5 Job 5: Year Wise Movies Count

Driver Class: (YearDriver.java)

```
package yearcount;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class YearDriver {
    public static void main(String[] args) throws Exception {
        if (args.length != 2) {
            System.err.println("Usage: YearDriver <input> <output>");
            System.exit(-1);
        }
        Configuration conf = new Configuration();
        Job job = Job.getInstance(conf, "Year Wise Title Count");
        job.setJarByClass(YearDriver.class);
        job.setMapperClass(YearMapper.class);
        job.setReducerClass(YearReducer.class);
        job.setMapOutputKeyClass(Text.class);
        job.setMapOutputValueClass(IntWritable.class);
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(IntWritable.class);
        FileInputFormat.addInputPath(job, new Path(args[0]));
        FileOutputFormat.setOutputPath(job, new Path(args[1]));
        System.exit(job.waitForCompletion(true) ? 0 : 1);
    }
}
```

Mapper Class: (YearMapper.java)

```
package yearcount;

import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class YearMapper
    extends Mapper<LongWritable, Text, Text, IntWritable> {
    private final Text yearKey = new Text();
    private final IntWritable one = new IntWritable(1);

    @Override
    public void map(LongWritable key, Text value, Context context)
        throws IOException, InterruptedException {
        String line = value.toString();
        // Skip header
        if (line.startsWith("Title_ID")) {
            return;
        }
        String[] fields = line.split("\t", -1);
        // Need at least 4 columns
        if (fields.length < 4) {
```

```

        return;
    }
    // Column 3 = Year
    String year = fields[3].trim();
    if (year.equals("\\N") || year.equals("Unknown")
        || year.isEmpty()) {
        return;
    }
    try {
        Integer.parseInt(year);
        yearKey.set(year);
        context.write(yearKey, one);
    } catch (NumberFormatException e) {
        // Ignore invalid years
    }
}
}
}

```

Reducer Class: (YearReducer.java)

```

package yearcount;
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class YearReducer
    extends Reducer<Text, IntWritable, Text, IntWritable> {
    private final IntWritable result = new IntWritable();

    @Override
    public void reduce(Text key, Iterable<IntWritable> values,
        Context context)
        throws IOException, InterruptedException {
        int total = 0;
        for (IntWritable value : values) {
            total += value.get();
        }
        result.set(total);
        context.write(key, result);
    }
}

```

EXECUTION

```
Select Administrator: Command Prompt
C:\Ashfak\IMDBMoviesAnalysis\yearcount>hadoop jar yearcount.jar YearDriver /imdb_movies/input/imdb_movies.txt /imdb_movies/output/yearcount
2026-09-13 17:49:59,673 INFO client.RPCProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 17:50:01,015 INFO mapreduce.jobresourceuploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-13 17:50:01,050 INFO mapreduce.jobresourceuploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Sudipta/-staging/job_1789296062547_0005
2026-09-13 17:50:01,540 INFO input.FileInputFormat: Total input files to process : 1
2026-09-13 17:50:01,727 INFO mapreduce.jobsubmitter: number of splits:2
2026-09-13 17:50:02,057 INFO mapreduce.jobsubmitter: Submitting tokens for job: job_1789296062547_0005
2026-09-13 17:50:02,061 INFO mapreduce.jobsubmitter: Executing with tokens: []
2026-09-13 17:50:02,428 INFO conf.Configuration: resource-types.xml not found
2026-09-13 17:50:02,429 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-13 17:50:02,588 INFO impl.YarnClientImpl: Submitted application application_1789296062547_0005
2026-09-13 17:50:02,682 INFO mapreduce.job: The url to track the job: http://DESKTOP-AD70200Q:8088/proxy/application_1789296062547_0005/
2026-09-13 17:50:02,685 INFO mapreduce.job: Running job: job_1789296062547_0005
2026-09-13 17:50:17,356 INFO mapreduce.job: Job job_1789296062547_0005 running in uber mode : false
2026-09-13 17:50:17,356 INFO mapreduce.job: map 0% reduce 0%
2026-09-13 17:50:33,980 INFO mapreduce.job: map 50% reduce 0%
2026-09-13 17:50:37,033 INFO mapreduce.job: map 100% reduce 0%
2026-09-13 17:50:47,322 INFO mapreduce.job: map 100% reduce 100%
2026-09-13 17:50:49,457 INFO mapreduce.job: Job job_1789296062547_0005 completed successfully
2026-09-13 17:50:49,911 INFO mapreduce.job: Counters: 55
  File System Counters
    FILE: Number of bytes read=21628438
    FILE: Number of bytes written=43974151
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=172940614
    HDFS: Number of bytes written=1500
    HDFS: Number of read operations=11
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=2
    HDFS: Number of bytes read erasure-coded=0
  Job Counters
    Killed map tasks=1
    Launched map tasks=2
    Launched reduce tasks=1
    Data-local map tasks=2
    Total time spent by all maps in occupied slots (ms)=31064
    Total time spent by all reducers in occupied slots (ms)=10535
    Total time spent by all map tasks (ms)=31064
    Total time spent by all reduce tasks (ms)=10535
    Total vcore-milliseconds taken by all map tasks=31064
    Total vcore-milliseconds taken by all reduce tasks=10535
    Total megabyte-milliseconds taken by all map tasks=31809536
    Total megabyte-milliseconds taken by all reduce tasks=10707840
```

execution1

Map-Reduce Framework

```
Map input records=2000001
Map output records=2000000
Map output bytes=17628432
Map output materialized bytes=21628444
Input split bytes=240
Combine input records=0
Combine output records=0
Reduce input groups=151
Reduce shuffle bytes=21628444
Reduce input records=2000000
Reduce output records=151
Spilled Records=4000000
Shuffled Maps =2
Failed Shuffles=0
Merged Map outputs=2
GC time elapsed (ms)=493
CPU time spent (ms)=16275
Physical memory (bytes) snapshot=1178963968
Virtual memory (bytes) snapshot=1474457600
Total committed heap usage (bytes)=963641344
Peak Map Physical memory (bytes)=497491968
Peak Map Virtual memory (bytes)=569819136
Peak Reduce Physical memory (bytes)=248569856
Peak Reduce Virtual memory (bytes)=338735104

Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0

File Input Format Counters
  Bytes Read=172940614
File Output Format Counters
  Bytes Written=1500
```

C:\Ashfak\IMDBMoviesAnalysis\yearcount>

Execution2

OUTPUT

```
Select Administrator: Command Prompt
C:\shf\AI\1000MoviesAnalysis\yearcount>hdfs -cat /adb_movies/output/yearcount/part-r-00000
173866
1874 1
1876 11
1881 21
1882 2
1887 1
1888 6
1889 2
1890 6
1891 13
1892 18
1893 7
1894 182
1895 111
1896 868
1897 1125
1898 1774
1899 1886
1900 1860
1901 1767
1902 1883
1903 2545
1904 1796
1905 1695
1906 1845
1907 2535
1908 4292
1909 4676
1910 5191
1911 6437
1912 6606
1913 9363
1914 9609
1915 8161
1916 6548
1917 5141
1918 3962
1919 3871
1920 4268
1921 3623
1922 3251
1923 2748
1924 2759
1925 3843
1926 3148
1927 3299
1928 3283
1929 3515
```

output1

```
Select Administrator: Command Prompt
1929 3515
1930 3064
1931 3163
1932 2957
1933 2883
1934 2886
1935 2884
1936 3245
1937 3268
1938 3137
1939 2902
1940 2548
1941 2566
1942 2482
1943 2235
1944 2040
1945 1990
1946 2230
1947 2413
1948 2556
1949 2750
1950 2976
1951 3005
1952 3227
1953 3497
1954 3578
1955 3771
1956 3946
1957 4204
1958 4279
1959 4456
1960 4830
1961 4919
1962 5021
1963 5291
1964 5733
1965 5847
1966 5774
1967 6160
1968 6498
1969 6519
1970 6926
1971 6868
1972 6689
1973 6614
1974 6474
1975 6916
1976 6821
1977 6809
```

output2

```
Select Administrator: Command Prompt
1977 6809
1978 7007
1979 7177
1980 7341
1981 7334
1982 7680
1983 8055
1984 8252
1985 8793
1986 8905
1987 9242
1988 9483
1989 9923
1990 10471
1991 10593
1992 11383
1993 11577
1994 12143
1995 13093
1996 13420
1997 14074
1998 15573
1999 16817
2000 18756
2001 20493
2002 22133
2003 25628
2004 30022
2005 34724
2006 37074
2007 39669
2008 43214
2009 49141
2010 54554
2011 59681
2012 63686
2013 65610
2014 60948
2015 25510
2016 21402
2017 24558
2018 33500
2019 73812
2020 82315
2021 85761
2022 81045
2023 79478
2024 78893
2025 72756
```

output3

3. Apache Pig Analysis

3.1 Job 1: Top 10 Highest-Rated Movies

The movies are sorted according to their ratings in descending order, and the top highest-rated movies are selected.

Script: (top5genres.pig)

```
infile = LOAD 'C:/Ashfak/data/imdb_movies.csv' USING PigStorage(',') AS (
    title_id:chararray, title:chararray, title_type:chararray, year_str:chararray,
    genres:chararray, runtime_str:chararray, rating_str:chararray, votes_str:chararray,
    director_id:chararray, director_name:chararray
);
clean_data = FILTER infile BY title_id != 'Title_ID' AND rating_str MATCHES '^[0-9]+(\\.[0-9]+)?$' AND votes_str MATCHES '^[0-9]+$';
typed_data = FOREACH clean_data GENERATE title_id, title, (float)rating_str AS rating, (int)votes_str AS votes, (int)year_str AS year;
credible_movies = FILTER typed_data BY votes >= 50;
ordered_data = ORDER credible_movies BY rating DESC;
top10 = LIMIT ordered_data 10;
STORE top10 INTO 'Top10RatedOutput' USING PigStorage('|');
```

EXECUTION

```
Select Administrator: Command Prompt
HadoopVersion  PigVersion  UserId  StartedAt      FinishedAt      Features
3.2.4         0.18.0         Sudipta  2026-09-13 18:30:11  2026-09-13 18:30:31  ORDER_BY,FILTER,LIMIT

Success!

Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReduceTime  Alias  Feature Outputs
job_local1097204110_0003  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  n/a  ORDER_BY,COMBINER
job_local152730123_0002  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  n/a  ordered_by_rating  SAMPLER
job_local2142808549_0001  6  0  0  n/a  n/a  n/a  0  0  0  0  infiles,valid_movies  MAP_ONLY
job_local682375754_0004  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  n/a  final_top10,ordered_by_rating  file:///C:/Ashfak/IMDBMoviesAnalysis/pig/Top10RatedOutput,

Input(s):
Successfully read 2000001 records from: "C:/Ashfak/data/imdb_movies.csv"

Output(s):
Successfully stored 10 records in: "file:///C:/Ashfak/IMDBMoviesAnalysis/pig/Top10RatedOutput"

Counters:
Total records written : 10
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local2142808549_0001  ->  job_local152730123_0002,
job_local152730123_0002  ->  job_local1097204110_0003,
job_local1097204110_0003  ->  job_local682375754_0004,
job_local682375754_0004

2026-09-13 18:30:31,619 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,627 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,627 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,639 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,643 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,643 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,659 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,661 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,672 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,679 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,683 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:30:31,693 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 399319 time(s).
2026-09-13 18:30:31,693 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-13 18:30:31,738 [main] INFO org.apache.pig.Main - Pig script completed in 24 seconds and 452 milliseconds (24452 ms)
```

Execution

OUTPUT

```
:\Ashfak\IMDBMoviesAnalysis\pig>type C:\Ashfak\IMDBMoviesAnalysis\pig\Top10RatedOutput\part-r-00000
t1433175|Romantic Weirdos and the Land of Oddz|2007|Drama|10.0|615
t30429361|Hug of Love 2|2023|Family|10.0|50
t39376578|Joy Awards 2026: Tuba B|Jy|Jk|Jst|Jn|2026|Short|10.0|174
t39305603|Joy Awards 2025: Tuba B|Jy|Jk|Jst|Jn|2026|Short|10.0|246
t0933381|Noel Gallagher: Sitting Here in Silence|2006|Music|9.9|111
t43367796|El Proyecto que Amenaza a M|~xico|2025|Documentary|9.9|80
t0205974|Golpe de Asa|1999|Short|9.9|90
t39144540|Red Sea Film Fest 2024|2025|Short|9.9|1091
t1954538|In Medias Res|2011|Thriller|9.9|770
t0188400|All About Lurleen|1992|Short|9.8|107

:\Ashfak\IMDBMoviesAnalysis\pig>
```

Output

3.2 Job 2: Movies Count per Genre

The total number of movies is calculated for each genre to identify how many movies belong to each genre.

Script: (genrecount.pig)

```
infile = LOAD 'C:/Ashfak/data/imdb_movies.csv' USING PigStorage(',') AS (
    title_id:chararray, title:chararray, title_type:chararray, year_str:chararray,
    genres:chararray, runtime_str:chararray, rating_str:chararray, votes_str:chararray,
    director_id:chararray, director_name:chararray
);

clean_data = FILTER infile BY title_id != 'Title_ID' AND genres IS NOT NULL AND genres != '';
flattened_genres = FOREACH clean_data GENERATE title_id, FLATTEN(TOKENIZE(genres, ',')) AS genre;
grouped_genre = GROUP flattened_genres BY TRIM(genre);
genre_count = FOREACH grouped_genre GENERATE group AS genre, COUNT(flattened_genres) AS total_movies;
ordered_count = ORDER genre_count BY total_movies DESC;
STORE ordered_count INTO 'GenreCountOutput' USING PigStorage('|');
```

EXECUTION

```
Select Administrator: Command Prompt
Success!

Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReduceTime  Alias  Feature Outputs
job_local1590748351_0002  1  1  n/a  n/a  n/a  n/a  n/a  n/a  ordered_count  SAMPLER  file:///C:/Ashfak/IMDBMoviesAnalysis/pig/GenreCountOutput,
job_local1212268652_0003  1  1  n/a  n/a  n/a  n/a  n/a  n/a  ordered_count  ORDER_BY  clean_data,flattened_genres,genre_count,grouped_genre,infiles
job_local1989876680_0001  6  1  n/a  n/a  n/a  n/a  n/a  n/a  clean_data,flattened_genres,genre_count,grouped_genre,infiles  GROUP_BY,COMBINER

Input(s):
Successfully read 2000001 records from: "C:/Ashfak/data/imdb_movies.csv"

Output(s):
Successfully stored 1088 records in: "file:///C:/Ashfak/IMDBMoviesAnalysis/pig/GenreCountOutput"

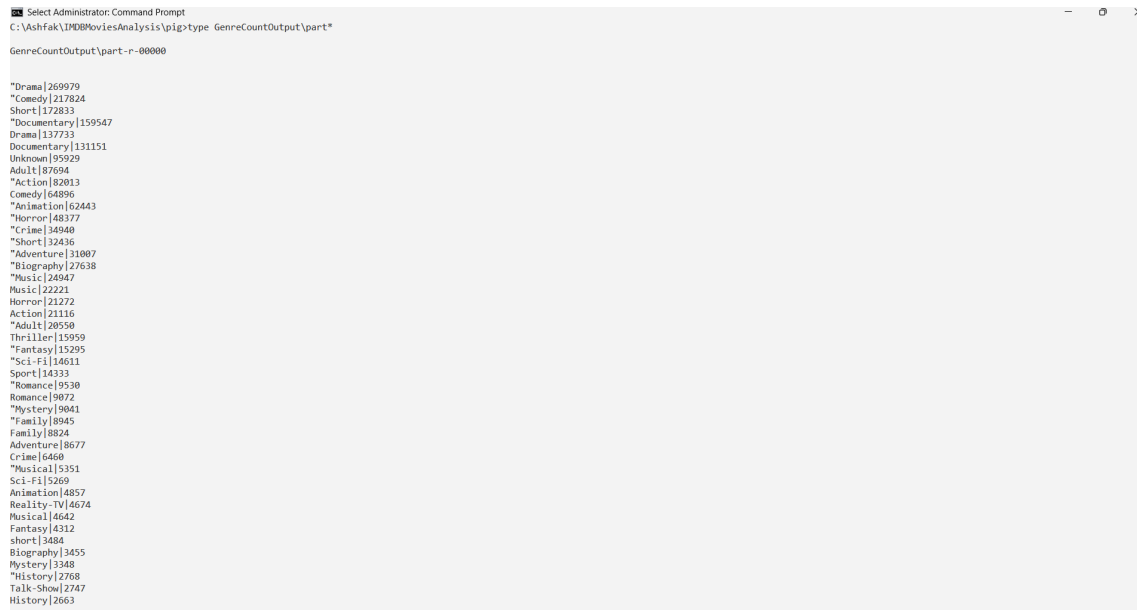
Counters:
Total records written : 1088
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1989876680_0001 -> job_local1590748351_0002,
job_local1590748351_0002 -> job_local1212268652_0003,
job_local1212268652_0003

2026-09-13 18:50:04,021 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:50:04,030 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:50:04,034 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:50:04,051 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:50:04,056 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:50:04,058 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:50:04,067 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:50:04,069 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:50:04,071 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:50:04,075 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-13 18:50:04,131 [main] INFO org.apache.pig.Main - Pig script completed in 23 seconds and 482 milliseconds (23482 ms)
```

Execution

OUTPUT



```
Select Administrator: Command Prompt
C:\Ashfak\IMDB\MoviesAnalysis\pig>type GenreCountOutput\part*
GenreCountOutput\part-r--00000

Drama|269979
Comedy|217824
Short|172833
Documentary|159547
Drama|137733
Documentary|131151
Unknown|95929
Adult|87694
Action|82013
Comedy|64896
Animation|62443
Horror|48377
Crime|34940
Short|32436
Adventure|31007
Biography|27638
Music|24947
Music|22221
Horror|21272
Action|21116
Adult|20550
Thriller|15959
Fantasy|15295
Sci-Fi|14611
Sport|14333
Romance|9530
Romance|9072
Mystery|9041
Family|8945
Family|8824
Adventure|8677
Crime|6460
Musical|5351
Sci-Fi|5269
Animation|4857
Reality-TV|4674
Musical|4642
Fantasy|4312
Short|3484
Biography|3455
Mystery|3348
History|2768
Talk-Show|2747
History|2663
```

Output

3.3 Job 3: Average Rating per Genre

The average rating of movies is calculated for each year to compare movie ratings across different years.

Script: (genreavg.pig)

```
infile = LOAD 'C:/Ashfak/data/imdb_movies.csv' USING PigStorage(',') AS (
    title_id:chararray, title:chararray, title_type:chararray, year_str:chararray,
    genres:chararray, runtime_str:chararray, rating_str:chararray, votes_str:chararray,
    director_id:chararray, director_name:chararray
);

clean_data = FILTER infile BY title_id != 'Title_ID' AND rating_str MATCHES '^[0-9]+(\\.[0-9]+)?$' AND genres IS NOT NULL;
typed_data = FOREACH clean_data GENERATE (float)rating_str AS rating, FLATTEN(TOKENIZE(genres, ',')) AS genre;
grouped_genre = GROUP typed_data BY TRIM(genre);
genre_avg = FOREACH grouped_genre GENERATE group AS genre, ROUND_TO(AVG(typed_data.rating), 2) AS avg_rating, COUNT(typed_data) AS
    total_movies;

ordered_avg = ORDER genre_avg BY avg_rating DESC;
STORE ordered_avg INTO 'GenreAvgOutput' USING PigStorage('|');
```

EXECUTION

```
Select Administrator: Command Prompt
success!

ob Stats (time in seconds):
obId    Maps    Reduces MaxMapTime    MinMapTime    AvgMapTime    MedianMapTime    MaxReduceTime    MinReduceTime    AvgReduceTime    MedianReduceTime    Alias    Feature Outputs
ob_local11850665402_0001    6    1    n/a    n/a    n/a    n/a    n/a    n/a    clean_data,genre_avg,grouped_genre,infiles,typed_data    GROUP_BY,COMBINER
ob_local1255382974_0003    1    1    n/a    n/a    n/a    n/a    n/a    ordered_avg    ORDER_BY    file:///C:/Ashfak/IMDBMoviesAnalysis/pig/GenreAvgOutput,
ob_local1265427801_0002    1    1    n/a    n/a    n/a    n/a    n/a    ordered_avg    SAMPLER

nput(s):
successfully read 2000001 records from: "C:/Ashfak/data/imdb_movies.csv"

otput(s):
successfully stored 861 records in: "file:///C:/Ashfak/IMDBMoviesAnalysis/pig/GenreAvgOutput"

ounters:
total records written : 861
total bytes written : 0
pillable Memory Manager spill count : 0
total bags proactively spilled: 0
total records proactively spilled: 0

ob DAG:
ob_local11850665402_0001    ->    job_local1265427801_0002,
ob_local1265427801_0002    ->    job_local1255382974_0003,
ob_local1255382974_0003

026-09-13 18:53:18,342 [main] WARN    org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
026-09-13 18:53:18,342 [main] WARN    org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
026-09-13 18:53:18,342 [main] WARN    org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
026-09-13 18:53:18,353 [main] WARN    org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
026-09-13 18:53:18,353 [main] WARN    org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
026-09-13 18:53:18,358 [main] WARN    org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
026-09-13 18:53:18,360 [main] WARN    org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
026-09-13 18:53:18,368 [main] WARN    org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
026-09-13 18:53:18,370 [main] WARN    org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
026-09-13 18:53:18,372 [main] INFO    org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
026-09-13 18:53:18,419 [main] INFO    org.apache.pig.Main - Pig script completed in 23 seconds and 260 milliseconds (23260 ms)
```

Execution

OUTPUT

```
Select Administrator: Command Prompt

C:\Ashfak\IMDBMoviesAnalysis\pig>type GenreAvgOutput\part*

GenreAvgOutput\part-r-000000

Pigasand"|2026.0|1
Who Always Goes Braless. - Asakaze Yui"|2026.0|1
Moe!"|2026.0|1
Resulting in an NTR (Netorare) Scenario"|2026.0|1
Judith & Tineke Steenbrink - Full Concert"|2026.0|1
Melody Hina Marks"|2026.0|1
and She Sucked and Swallowed My Load Over and Over Again Until I Couldn't Get It to Get Hard Anymore. Nijimura Yumi"|2026.0|1
Slutty Wife Gets Creampied in a Crab-Legged Cowgirl Position."|2026.0|1
Elsewhere"|2026.0|1
BEST 4"|2026.0|1
and Makes Her AV Debut. Track and Field Girl (21)"|2026.0|1
2023"|2026.0|1
25 Years Old"|2026.0|1
Hanging Restraint"|2026.0|1
and She Comes All Over the Place!"|2026.0|1
Mia Kouki"|2026.0|1
and Years"|2026.0|1
Daisies"|2026.0|1
Boca"|2026.0|1
& Purder"|2026.0|1
and Fashionable Artist!"|2026.0|1
Hidden Truth"|2026.0|1
4 Aspiring Idols Who Were Played to Provide Sexual Services After Being Soaked in It"|2026.0|1
Asking for Unprotected Sex!"|2026.0|1
Downtown Dattel 37 June Lovejoy"|2026.0|1
divorciada 2"|2025.0|1
and the Day After Tomorrow"|2025.0|1
Revised Edition"|2025.0|1
Lasiadora"|2025.0|1
Aliens and UFOs"|2025.0|1
Making Her Sensitive and Quick to Cum and a Premature Ejaculation Constitution"|2025.0|1
Crete"|2025.0|1
Death"|2025.0|1
an Active Masturbation Pet High School Girl Who's Seen Men Ejaculating Too Much"|2025.0|1
Where You Can Have as Much Sex as You Want! - Melody Hina Marks"|2025.0|1
Lead Development of High School Girls in Their School Life."|2025.0|1
K-Cup Breasts Massaged in Super-Rich Creampie Three-Shot"|2025.0|1
Cumming a Total of 10 Times in a Row! Kuroshima Rei"|2025.0|1
Facial Cumshot Super Collection Best Uncut 1270Min"|2025.0|1
Flat-Skinned Crossdresser Eno Enoki's AV Debut"|2025.0|1
Very Bad Road Trip"|2025.0|1
les Preyres"|2025.0|1
tiPealea..."|2025.0|1
```

Output

3.4 Job 4: Average Rating & Release Count per Year

Script: (yearavg.pig)

```
infile = LOAD 'C:/Ashfak/data/imdb_movies.csv' USING PigStorage(',') AS (
    title_id:chararray, title:chararray, title_type:chararray, year_str:chararray,
    genres:chararray, runtime_str:chararray, rating_str:chararray, votes_str:chararray,
    director_id:chararray, director_name:chararray
);
clean_data = FILTER infile BY title_id != 'Title_ID' AND year_str MATCHES '^[0-9]{4}$' AND rating_str MATCHES '^[0-9]+(\\. [0-9]+)?$';
typed_data = FOREACH clean_data GENERATE (int)year_str AS year, (float)rating_str AS rating;
grouped_year = GROUP typed_data BY year;
year_summary = FOREACH grouped_year GENERATE group AS year, COUNT(typed_data) AS movie_count, ROUND_TO(AVG(typed_data.rating), 2) AS
    avg_rating;
ordered_year = ORDER year_summary BY year DESC;
STORE ordered_year INTO 'YearAvgOutput' USING PigStorage('|');
```

EXECUTION

```
Select Administrator: Command Prompt
Success!

Job Stats (time in seconds):
JobId  Maps    Reduces  MaxMapTime    MinMapTime    AvgMapTime    MedianMapTime    MaxReduceTime    MinReduceTime    AvgReduceTime    MedianReduceTime    Alias    Feature Outputs
job_local202149297_0002  1      1      n/a      n/a      n/a      n/a      n/a      n/a      n/a      n/a      ordered_year  SAMPLER
job_local46620700_0001  6      1      n/a      n/a      n/a      n/a      n/a      n/a      n/a      n/a      clean_data,grouped_year,infiles,typed_data,year_summary  GROUP_BY,COMBINER
job_local64788502_0003  1      1      n/a      n/a      n/a      n/a      n/a      n/a      n/a      n/a      ordered_year  ORDER_BY      file:///C:/Ashfak/IMDBMoviesAnalysis/pig/YearAvgOutput,

Input(s):
Successfully read 2000001 records from: "C:/Ashfak/data/imdb_movies.csv"

Output(s):
Successfully stored 148 records in: "file:///C:/Ashfak/IMDBMoviesAnalysis/pig/YearAvgOutput"

Counters:
Total records written : 148
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local46620700_0001 -> job_local202149297_0002,
job_local202149297_0002 -> job_local64788502_0003,
job_local64788502_0003

2026-09-13 18:57:22,434 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:57:22,437 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:57:22,437 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:57:22,453 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:57:22,453 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:57:22,453 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:57:22,463 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:57:22,465 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:57:22,469 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 18:57:22,478 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-13 18:57:22,514 [main] INFO org.apache.pig.Main - Pig script completed in 24 seconds and 440 milliseconds (24440 ms)

C:\Ashfak\IMDBMoviesAnalysis\pig>
```

Execution

OUTPUT

```
Select Administrator Command Prompt
':\usb\fs1\HDD\NewerMedia\ps\ps1\ps1-type Year\fsOutput\ps1.r'
fsOutput\ps1.r - 00000

2012|11|08.0
2010|11|17
2020|11|08.91
2010|11|17.33
2027|16|713.24
2020|11|17.00.42
2005|10|002|10.15
2016|10|042|9.37
2023|08|766|10.31
2027|00|19.35.54
2021|74|736.15
2007|20|57.7.43
2019|54|333|7.78
2018|70|651.6.79
2017|27|4681.6.89
2016|17|0811.18
2015|10|080|12.29
2014|25|062|12.57
2013|16|2077|12.34
2012|16|070|11.06
2011|16|073|12.16
2010|10|020|12.63
2009|17|770|12.47
2008|16|122|11.38
2007|11|068|11.26
2006|11|015|11.43
2005|10|047|11.52
2004|25|099|12.07
2003|12|040|12.91
2002|16|701|13.52
2001|17|416|12.78
2000|16|031|13.31
1999|14|046|13.2
1998|11|156|13.4
1997|11|156|14.23
1996|11|413|14.26
1995|11|136|14.11
1994|10|027|14.42
1993|09|003|15.28
1992|10|772|14.81
1991|10|771|16.1
1990|08|21|15.75
1989|15|471|17.27
```

Output

```

C:\ Select Administrator: Command Prompt
1929|2921|9.59
1928|2806|10.84
1927|2788|10.98
1926|2592|11.62
1925|2538|9.52
1924|2341|9.21
1923|2333|8.04
1922|2729|7.7
1921|3308|6.21
1920|3759|5.51
1919|3473|6.06
1918|3496|6.99
1917|4450|5.27
1916|5649|3.22
1915|7106|2.34
1914|7759|2.56
1913|8102|3.61
1912|7053|1.46
1911|5657|1.14
1910|4555|0.87
1909|4254|0.7
1908|3853|0.61
1907|2242|0.61
1906|1603|0.99
1905|1335|0.69
1904|1504|0.58
1903|2219|0.62
1902|1564|0.42
1901|1453|0.46
1900|1623|0.53
1899|1478|0.5
1898|1480|0.71
1897|1102|1.06
1896|686|1.13
1895|84|1.72
1894|81|2.47
1893|7|1.74
1892|9|2.86
1891|9|2.9
1890|2|1.0
1889|1|1.0
1888|5|1.72
1887|1|1.0
1882|2|1.0
1881|21|4.57
1878|30|1.0

YearAvgOutput\ .part-r-00000.crc

```

Output

3.5 Job 5: Top Directors by Movie Count

Script: (directorcount.pig)

```
infile = LOAD 'C:/Ashfak/data/imdb_movies.csv' USING PigStorage(',') AS (
    title_id:chararray, title:chararray, title_type:chararray, year_str:chararray,
    genres:chararray, runtime_str:chararray, rating_str:chararray, votes_str:chararray,
    director_id:chararray, director_name:chararray
);
clean_data = FILTER infile BY title_id != 'Title_ID' AND director_name IS NOT NULL AND director_name != '' AND director_name != '\\\\';
grouped_director = GROUP clean_data BY director_name;
director_summary = FOREACH grouped_director GENERATE group AS director_name, COUNT(clean_data) AS total_movies;
ordered_directors = ORDER director_summary BY total_movies DESC;
top_directors = LIMIT ordered_directors 20;
STORE top_directors INTO 'DirectorCountOutput' USING PigStorage('|');
```

EXECUTION

```
Select Administrator: Command Prompt
Success!

Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReduceTime  Alias  Feature Outputs
job_local1578140241_0002  1  1  n/a  n/a  n/a  n/a  n/a  n/a  ordered_directors  SAMPLER
job_local1594569418_0003  1  1  n/a  n/a  n/a  n/a  n/a  n/a  ordered_directors  ORDER_BY_COMBINER
job_local1605131677_0004  1  1  n/a  n/a  n/a  n/a  n/a  n/a  ordered_directors  file:///C:/Ashfak/IMDBMoviesAnalysis/pig/DirectorCountOutput,
job_local12075160789_0001  6  1  n/a  n/a  n/a  n/a  n/a  n/a  clean_data,director_summary,grouped_director,infiles  GROUP_BY_COMBINER

Input(s):
Successfully read 2000001 records from: "C:/Ashfak/data/imdb_movies.csv"

Output(s):
Successfully stored 20 records in: "file:///C:/Ashfak/IMDBMoviesAnalysis/pig/DirectorCountOutput"

Counters:
Total records written : 20
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local12075160789_0001  ->  job_local1578140241_0002,
job_local1578140241_0002  ->  job_local1594569418_0003,
job_local1594569418_0003  ->  job_local1605131677_0004,
job_local1605131677_0004

2026-09-13 19:00:48,460 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:00:48,468 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:00:48,473 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:00:48,489 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:00:48,495 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:00:48,503 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:00:48,511 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:00:48,515 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:00:48,515 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:00:48,526 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:00:48,526 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:00:48,528 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:00:48,528 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:00:48,542 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-13 19:00:48,606 [main] INFO org.apache.pig.Main - Pig script completed in 31 seconds and 175 milliseconds (31175 ms)
```

Execution

OUTPUT

```
C:\Ashfak\IMDBMoviesAnalysis\pig>type DirectorCountOutput\part*
```

```
DirectorCountOutput\part-r-00000
```

```
Unknown|262470
```

```
0|184246
```

```
9|3370
```

```
8|3264
```

```
10|3220
```

```
7|3183
```

```
11|2968
```

```
12|2740
```

```
13|2635
```

```
14|2376
```

```
6|2354
```

```
15|2212
```

```
16|2085
```

```
18|1874
```

```
17|1872
```

```
19|1680
```

```
20|1615
```

```
21|1524
```

```
22|1451
```

```
5|1319
```

```
DirectorCountOutput\part-r-00000.crc
```

```
ccac01
```

Output

3.6 Job 6: Content Count by Title Type

Script: (typecount.pig)

```
infile = LOAD 'C:/Ashfak/data/imdb_movies.csv' USING PigStorage(',') AS (
    title_id:chararray, title:chararray, title_type:chararray, year_str:chararray,
    genres:chararray, runtime_str:chararray, rating_str:chararray, votes_str:chararray,
    director_id:chararray, director_name:chararray
);
clean_data = FILTER infile BY title_id != 'Title_ID' AND title_type IS NOT NULL;
grouped_type = GROUP clean_data BY title_type;
type_summary = FOREACH grouped_type GENERATE group AS title_type, COUNT(clean_data) AS total_titles;
ordered_type = ORDER type_summary BY total_titles DESC;
STORE ordered_type INTO 'TypeCountOutput' USING PigStorage('|');
```

EXECUTION

```
Select Administrator: Command Prompt
Success!

Job Stats (time in seconds):
JobId   Maps   Reduces MaxMapTime   MinMapTime   AvgMapTime   MedianMapTime   MaxReduceTime   MinReduceTime   AvgReduceTime   MedianReduceTime   Alias   Feature Outputs
job_local1805371738_0001   6         0         n/a         n/a         n/a         0         0         0         clean_data,infiles,popular_hits,typed_data   MAP_ONLY
job_local1235722055_0004   1         1         n/a         n/a         n/a         n/a         n/a         n/a         ordered_hits   file:///C:/Ashfak/IMDBMoviesAnalysis/pig/HighRatedPopularOutput,
job_local1882535059_0003   1         1         n/a         n/a         n/a         n/a         n/a         n/a         ordered_hits   ORDER_BY_COMBINER
job_local684680715_0002   1         1         n/a         n/a         n/a         n/a         n/a         n/a         ordered_hits   SAMPLER

Input(s):
Successfully read 2000001 records from: "C:/Ashfak/data/imdb_movies.csv"

Output(s):
Successfully stored 20 records in: "file:///C:/Ashfak/IMDBMoviesAnalysis/pig/HighRatedPopularOutput"

Counters:
Total records written : 20
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1805371738_0001   ->   job_local684680715_0002,
job_local684680715_0002 ->   job_local1882535059_0003,
job_local1882535059_0003   ->   job_local1235722055_0004,
job_local1235722055_0004

2026-09-13 19:14:43,230 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,237 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,239 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,245 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,245 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,245 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,257 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,258 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,260 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,266 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,268 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,273 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:14:43,282 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 16613 time(s).
2026-09-13 19:14:43,282 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-13 19:14:43,337 [main] INFO org.apache.pig.Main - Pig script completed in 17 seconds and 709 milliseconds (17709 ms)
```

Execution

OUTPUT

```
C:\Ashfak\IMDBMoviesAnalysis\pig>type HighRatedPopularOutput\part*

HighRatedPopularOutput\part-r-00000

The Shawshank Redemption|9.3|3233087|1994
Batman Begins|8.2|1741893|2005
American Beauty|8.3|1287445|1999
One Flew Over the Cuckoo's Nest|8.6|1165820|1975
Raiders of the Lost Ark|8.4|1119851|1981
Requiem for a Dream|8.3|983780|2000
Indiana Jones and the Last Crusade|8.2|870773|1989
Gran Torino|8.1|869501|2008
There Will Be Blood|8.2|718146|2007
Rain Man|8.0|589161|1988
The Kashmir Files|8.5|579811|2022
The Help|8.1|532629|2011
The Exorcist|8.1|504094|1973
Monty Python's Life of Brian|8.0|435717|1979
The Hunt|8.3|417446|2012
Scent of a Woman|8.0|375708|1992
Magnolia|8.0|352787|1999
A Separation|8.3|280196|2011
Bicycle Thieves|8.2|193133|1948
Network|8.0|187504|1976

HighRatedPopularOutput\.part-r-00000.crc

crcxmV∞[i.~
C:\Ashfak\IMDBMoviesAnalysis\pig>
```

Output

3.7 Job 7: Bottom 20 Lowest-Rated Titles

Script: (bottomrated.pig)

```
infile = LOAD 'C:/Ashfak/data/imdb_movies.csv' USING PigStorage(',') AS (
    title_id:chararray, title:chararray, title_type:chararray, year_str:chararray,
    genres:chararray, runtime_str:chararray, rating_str:chararray, votes_str:chararray,
    director_id:chararray, director_name:chararray
);
-- Clean data and cast types safely
clean_data = FILTER infile BY
    title_id != 'Title_ID' AND
    rating_str MATCHES '^[0-9]+(\\. [0-9]+)?$' AND
    votes_str MATCHES '^[0-9]+$';
typed_data = FOREACH clean_data GENERATE
    title_id,
    title,
    (int)year_str AS year,
    genres,
    (float)rating_str AS rating,
    (int)votes_str AS votes;
-- Ensure minimum vote threshold for credibility
credible_movies = FILTER typed_data BY votes >= 50;
-- Sort by rating ascending (lowest first)
ordered_bottom = ORDER credible_movies BY rating ASC, votes DESC;
bottom20_rated = LIMIT ordered_bottom 20;
STORE bottom20_rated INTO 'Bottom20RatedOutput' USING PigStorage('|');
```

EXECUTION

```
Select Administrator: Command Prompt
Success!

Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReduceTime  Alias  Feature Outputs
job_local117611623_0004  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  n/a  ordered_bottom  file:///C:/Ashfak/IDDBMoviesAnalysis/pig/Bottom20RatedOutput,
job_local1468242486_0002  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  n/a  ordered_bottom  SAMPLER  Clean_data,credible_movies,infiles,typed_data  MAP_ONLY
job_local2020351085_0001  6  0  n/a  n/a  n/a  n/a  0  0  0  0  ordered_bottom  ORDER_BY,COMBINER
job_local1818631992_0003  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  n/a  ordered_bottom  ORDER_BY,COMBINER

Input(s):
Successfully read 2000001 records from: "C:/Ashfak/data/imdb_movies.csv"

Output(s):
Successfully stored 20 records in: "file:///C:/Ashfak/IDDBMoviesAnalysis/pig/Bottom20RatedOutput"

Counters:
Total records written : 20
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local2020351085_0001 -> job_local1468242486_0002,
job_local1468242486_0002 -> job_local1818631992_0003,
job_local1818631992_0003 -> job_local117611623_0004,
job_local117611623_0004

2026-09-13 19:07:47,565 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,584 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,586 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,595 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,597 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,600 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,604 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,606 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,608 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,618 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,621 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,621 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,622 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-13 19:07:47,625 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 16613 time(s).
2026-09-13 19:07:47,627 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-13 19:07:47,687 [main] INFO org.apache.pig.Main - Pig script completed in 24 seconds and 699 milliseconds (24699 ms)
```

Execution

OUTPUT

```
C:\Ashfak\IMDBMoviesAnalysis\pig>type Bottom20RatedOutput\part*  
Bottom20RatedOutput\part-r-00000  
  
tt13423846|321 Action|2020|Drama|1.0|10341  
tt37836776|Kadik|y Bogasi|2025|Comedy|1.0|7044  
tt1852818|Justin Bieber: Rise to Fame|2011|Documentary|1.0|2845  
tt15443254|Cumali Ceber 666|2022|Comedy|1.0|1190  
tt11529726|Laz Kit|2020|Comedy|1.0|963  
tt32125636|Donald Trump: Made in the USA|2024|Unknown|1.0|810  
tt22485534|Think Big and Kick Ass in Business and Life|2007|Unknown|1.0|436  
tt0334193|QIK2JG|2002|Short|1.0|294  
tt13232730|Hearts Are Trump|2020|Drama|1.0|256  
tt10642850|15 Temmuz destani|2019|News|1.0|217  
tt29868945|Dressing Like Meghan Markle|2023|Documentary|1.0|203  
tt0884106|Jocking Around|2007|Documentary|1.0|191  
tt1392266|Who's Watching Who?|2000|Comedy|1.0|182  
tt1872215|Tunnel Rendez-vous|2011|Comedy|1.0|155  
tt21938460|Trip to the Moon|2007|Short|1.0|148  
tt12280046|Queer & Fier(ce)|2019|Documentary|1.0|103  
tt0374100|Onnanoko no heya: O-teire namachuukei!!|1998|Unknown|1.0|67  
tt1914390|Toukyou doutei|2010|Short|1.0|66  
tt0310048|Reality|1969|Documentary|1.0|66  
tt0196388|Blood of the Undead: The Unwanted|1987|Short|1.0|59
```

Output

—End of Report—